✈ DGCA CPL / ATPL STUDY NOTES

CHAPTER 7

Automatic Direction Finder (ADF)

Radio Navigation & Aids — Ground Training Series

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1. Introduction

The **Automatic Direction Finder (ADF)** is airborne equipment used in conjunction with a ground-based **Non-Directional Beacon (NDB)** to provide navigation assistance and non-precision approach procedures at aerodromes.

Historical Status

ADF/NDB was due to be phased out in 2005 but **continues in use**. Many UK aerodromes still publish NDB instrument approach procedures, and at some aerodromes it remains the *only* instrument approach procedure available.

Key Advantage

ADF requires **no special ground equipment beyond the NDB** and is available on any frequency in the LF/MF band. It provides low-cost, widely available backup navigation.

2. Non-Directional Beacon (NDB)

The NDB is a **ground-based transmitter** that radiates **vertically polarized** radio signals equally in all directions — hence "non-directional."

- Operates in the **LF and MF bands** (190 – 1750 kHz)
- When an ADF is tuned to an NDB's frequency and the callsign is identified, the NDB's direction is indicated
- A **cone of silence** exists directly overhead the NDB — no signal received while directly overhead; cone diameter increases with aircraft height

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Cone of Silence

When the aircraft is directly overhead the NDB, no signal is received and the ADF needle will swing erratically. The diameter of the cone of silence **increases with altitude**.

3. Principle of Operation

ADF measures the bearing of an NDB **relative to the fore/aft axis of the aircraft**.

3.1 Loop Aerial

A voltage is generated in the vertical elements of a loop aerial due to the **phase difference** of the incoming wave across the loop. As the loop rotates, the induced voltage varies, reaching **zero when the loop is perpendicular to the radio wave** — this is the "null."

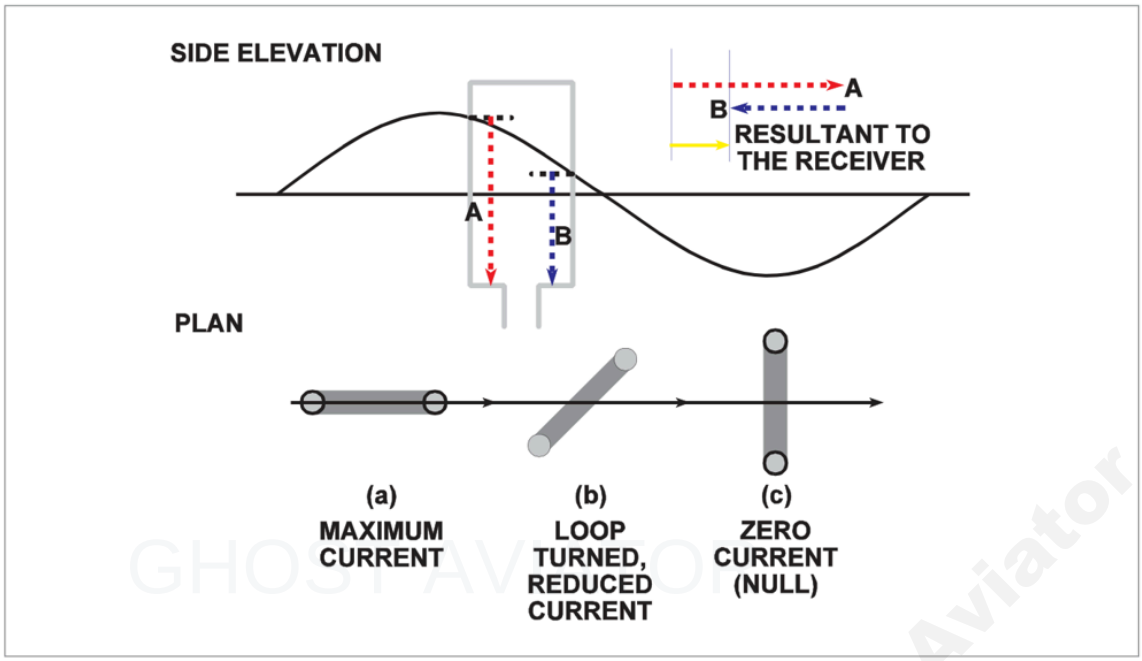


Figure 7.1 A Loop Aerial

Figure 7.1 — A loop aerial: the voltage induced by an incoming wave varies as the loop rotates, reaching null (zero) when the loop plane is perpendicular to the wave.

The loop produces a **figure-of-eight polar diagram** with two null positions. This creates a 180° ambiguity — two possible directions for the beacon.

3.2 Sense Aerial — Resolving Ambiguity

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problem.

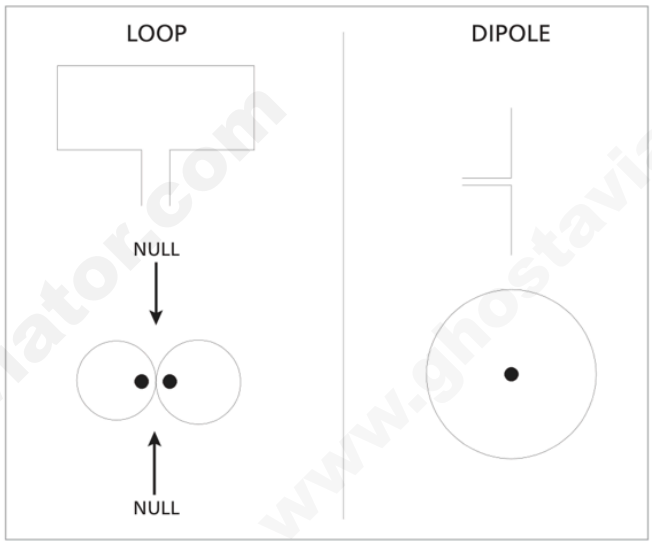


Figure 7.2 Polar diagrams of loop & dipole aerials

To resolve this ambiguity a simple dipole aerial, called a sense aerial, is added. The polar diagram of the sense aerial is circular. The currents generated are combined electronically as if the sense aerial was in the middle of the loop aerial (*Figure 7.3*). The relative signal strengths of the two signals are shown.

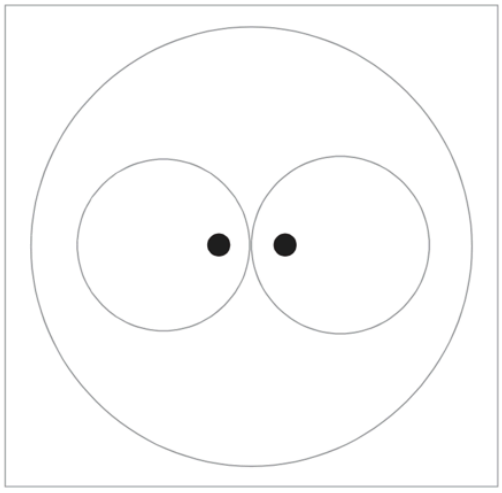


Figure 7.3

Figure 7.2 — Polar diagrams: loop aerial produces figure-of-eight with two nulls (ambiguous); adding a dipole sense aerial with circular pattern resolves ambiguity. Figure 7.3 shows the combined polar diagram with relative signal strengths.

To resolve the ambiguity, a simple **dipole (sense) aerial** is added. Its polar diagram is circular. The signals from loop and sense aerials are combined electronically (as if the sense aerial is at the centre of the loop). The result is a **CARDIOID** — a heart-shaped pattern with a *single* null.

3.3 Cardioid — Single Null

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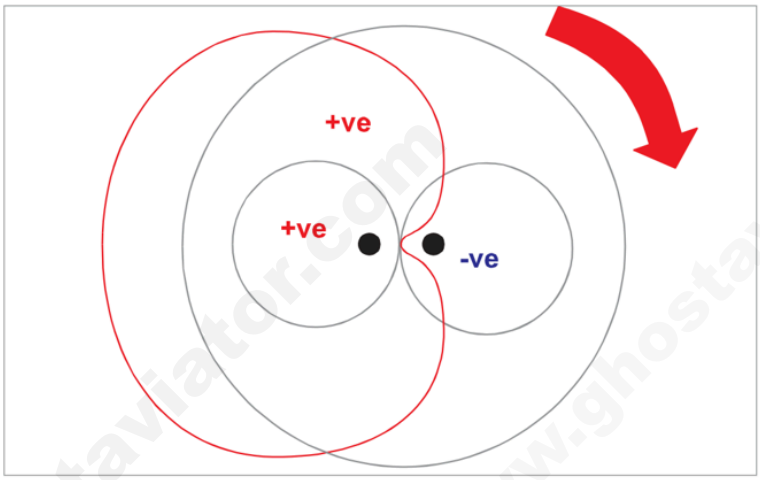


Figure 7.4

The resultant null with a single cardioid is not precise enough to meet the ICAO accuracy requirement of +/-5°. To improve the accuracy to meet the requirements, the polarity of the sense aerial is reversed to produce a right hand cardioid. Then by rapidly switching (about 120 Hz) between the two cardioids, the null is more precisely defined and hence the accuracy is improved.

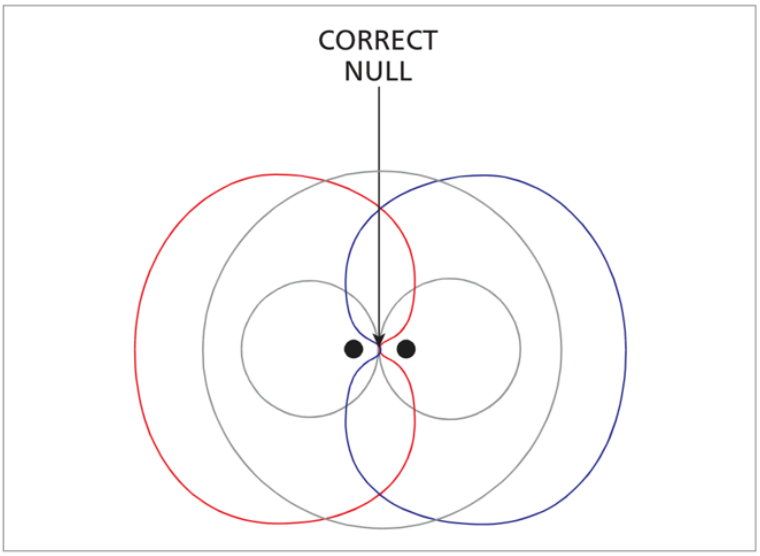


Figure 7.5

Figure 7.4 — Cardioid formed by combining loop + sense aerials (left cardioid shown). Figure 7.5 — By rapidly switching (~120 Hz) between left and right cardioids, the null is more precisely defined, meeting ICAO accuracy of ±5°.

Dual Cardioid Switching

A single cardioid null is **not precise enough** for ICAO ±5° accuracy. The polarity of the sense aerial is reversed to produce a right-hand cardioid. By **rapidly switching (~120 Hz)** between the two cardioids, the null is more precisely defined — meeting the ICAO accuracy requirement.

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3.4 Fixed Loop / Goniometer

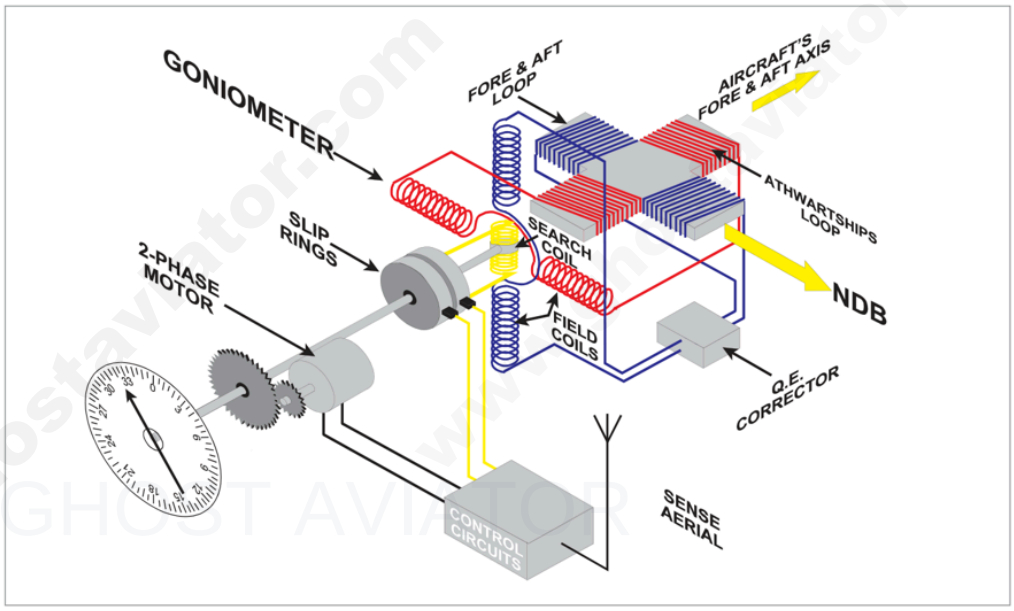


Figure 7.6 Fixed Loop ADF

Figure 7.6 — Fixed loop ADF system: a fixed four-element aerial (two fore/aft, two lateral) feeds a goniometer, which reproduces the electromagnetic field. A search coil in the goniometer detects the unambiguous bearing — eliminating the need for a rotating external loop.

In practice, a rotating loop outside the aircraft is not feasible. The ADF uses a **fixed four-element loop** (two fore/aft, two lateral) feeding a **goniometer** — a device that electronically reproduces the electromagnetic field and extracts the bearing using a search coil.

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4. Frequencies and Types of NDB

NDB Frequency Allocation

Allocated band: **190 – 1750 kHz** (LF and MF bands)
Most NDBs are found between **250 – 450 kHz** (surface wave propagation most reliable here).
Propagation mode: **Surface wave**

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Type	Range	Use	Availability
Locator (L)	**10 – 25 NM**	Airfield/runway approach procedures; co-located with ILS outer or middle marker	May only be available during published aerodrome hours
En Route NDB	**50 NM or more** (oceanic: hundreds of miles)	Homing, holding, en route and airways navigation	Continuous; listed in COMM section of AIP

```
NDB RANGE FORMULA

Range = 3 × √P(W)   — over water

Range = 2 × √P(W)   — over land

P = NDB transmitter power in Watts  |  Range in NM
```

5. Aircraft Equipment

The ADF system in the aircraft comprises:

- A **loop aerial** (fixed four-element)
- A **sense aerial**
- A **control unit** (frequency selector, ADF/ANT/BFO switches)
- A **receiver**
- A **display** (RBI or RMI)

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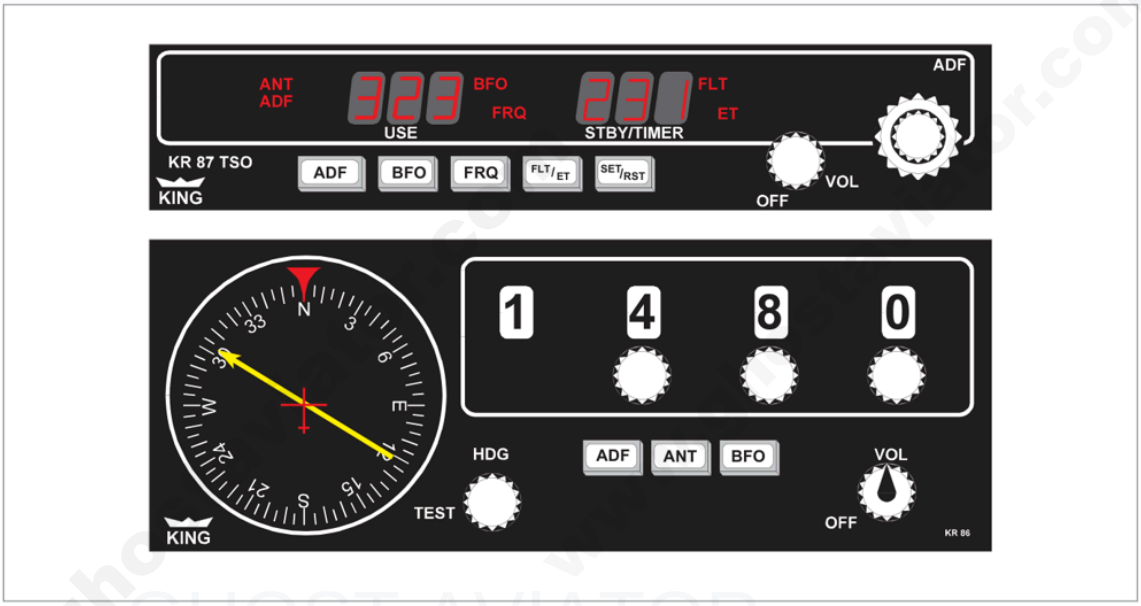


Figure 7.7 Two ADF Receivers

Figure 7.7 — Two ADF receivers: the control unit allows frequency selection and BFO (Beat Frequency Oscillator) selection, with ADF/ANT/BFO mode switch.

6. Emission Characteristics and BFO

NDBs have a **2 or 3 letter identification callsign**. There are two emission types:

N0NA1A

Unmodulated carrier (N0N) + interrupted unmodulated carrier (A1A)
A1A requires BFO to be ON to produce an audible tone for identification.

BFO ON for: Tuning, Identification, AND Monitoring

N0NA2A

Unmodulated carrier (N0N) + amplitude modulated carrier (A2A)
A2A can be heard on a normal receiver without BFO.

BFO ON for: Tuning only
BFO OFF for: Identification and Monitoring

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BFO Rules — Must Know for Exam

Emission	Tune	Identify	Monitor
N0NA1A	ON	ON	ON
N0NA2A	ON	OFF	OFF

Note: BFO may be labelled **TONE** or **TONE/VOICE** on some equipment.

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7. Presentation of Information

ADF information is presented on either a **Relative Bearing Indicator (RBI)** or a **Radio Magnetic Indicator (RMI)**. Both display **relative bearing**.



Figure 7.8 RBI

Figure 7.9 RMI

Figure 7.8 (left) — RBI: fixed 360° rose with 000° at nose; read relative bearing directly. Example: hdg 300°M, RBI 136° → magnetic bearing to NDB = 300° + 136° − 360° = 076°M. Figure 7.9 (right) — RMI: compass card rotates with aircraft heading; needle shows magnetic bearing directly.

Instrument	Compass Card	What Needle Shows	To Get Mag Bearing
RBI	Fixed — 000° always at nose	Relative bearing (angle from nose)	Heading + RBI reading (−360° if >360°)
RMI	Rotates with aircraft heading	Magnetic bearing directly	Read directly from under needle head

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QDM / QDR from ADF

- **Needle HEAD** → points TO the NDB = **QDM** (if RMI, magnetic heading to station)
- **Needle TAIL** → points FROM the NDB = **QDR** (magnetic bearing from station = radial)

8. Uses of the Non-Directional Beacon

- **En route navigational bearings**
- **Homing** to or flying from the NDB when maintaining airway centrelines
- **Holding** overhead at an assigned level in a racetrack pattern
- **Runway instrument approach procedures** (non-precision approach)

Plotting ADF Bearings

When converting an ADF bearing to a position line on a chart, variation must be applied **at the aircraft's position** (not the NDB). For long-range bearings, convergency between the aircraft and beacon meridians must also be applied.

9. Track Maintenance Using the RBI

9.1 Homing

Homing = maintaining 000° relative bearing (NDB dead ahead at all times). In zero wind, the aircraft flies a straight track inbound. In crosswind, the aircraft follows a *curved* track (the heading constantly changes to keep the needle at 000°).

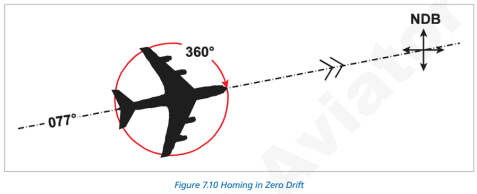


Figure 7.10 Homing in Zero Drift

Figure 7.10 — Homing in zero wind: aircraft heading 077°, RBI 000°. Straight track to NDB.

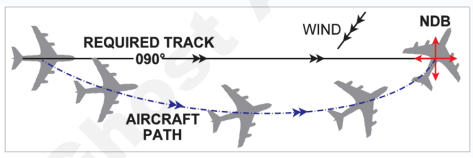


Figure 7.11 Homing Making no Allowance for Drift

Figure 7.11 — Homing with crosswind (no drift allowance): aircraft maintains 000° RBI but follows a curved track — arriving at NDB from downwind side.

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9.2 Tracking Inbound

To fly a *straight* track inbound, the pilot applies a **Wind Correction Angle (WCA)**:

Inbound Track Rule

Required Relative Bearing = **360° − WCA** (for starboard drift) or **WCA** (for port drift)

- Starboard drift anticipated → **Subtract** WCA from track → heading to fly = track − WCA; RBI reads 360° − (−WCA) = 360° + WCA?
  Simpler rule: **S**tarboard drift → **S**ubtract WCA from track heading
- Port drift anticipated → **Add** WCA to track heading (Plus for Port)

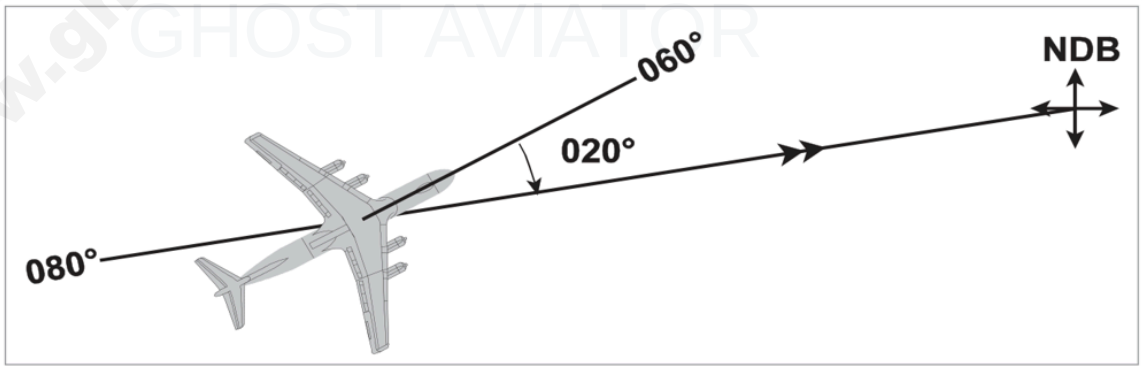


Figure 7.12

In *Figure 7.13*, 28° Port drift is anticipated, so this is added (Plus) to the track value. The aircraft is heading 108° with a relative bearing of 332°.

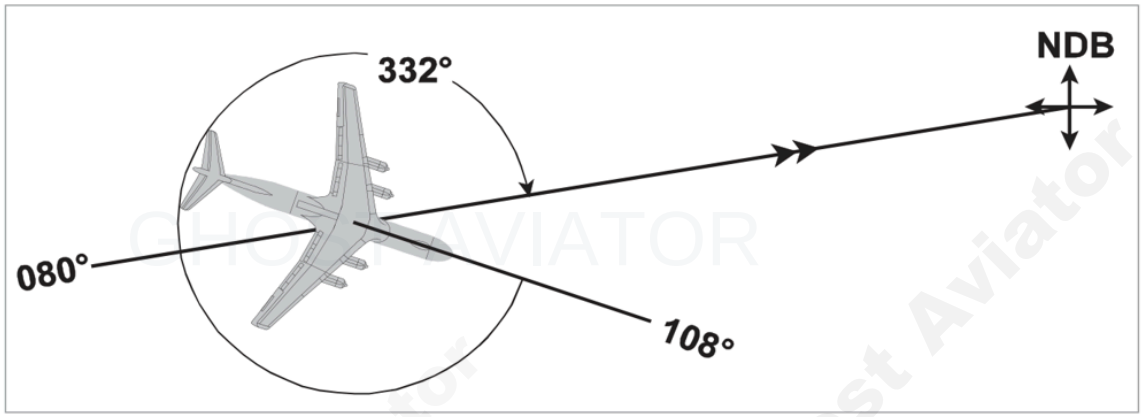


Figure 7.13

Figure 7.12 — 20° starboard drift: heading = track − 20°; aircraft heading 060°, RBI 020°. Figure 7.13 — 28° port drift: heading = track + 28°; aircraft heading 108°, RBI 332°.

9.3 Tracking Outbound

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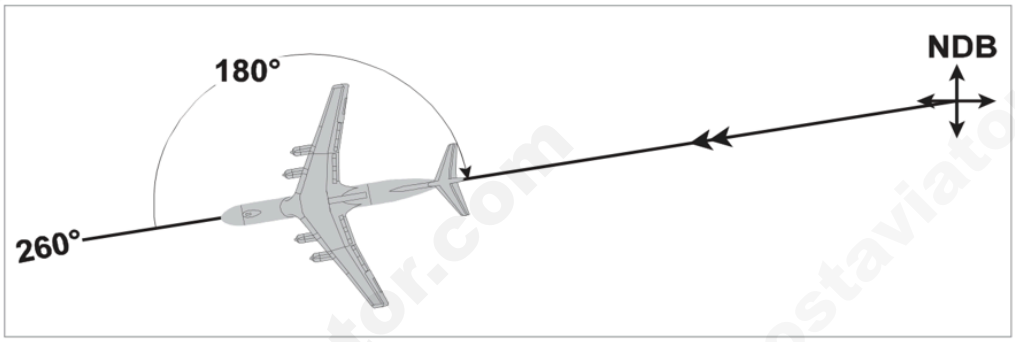


Figure 7.14

Figure 7.15 shows an aircraft maintaining a track of 100° in crosswind conditions where the drift is known. 23° of Starboard drift is anticipated, this is Subtracted from the track, therefore the heading is 077° with a relative bearing of 203° from the NDB.

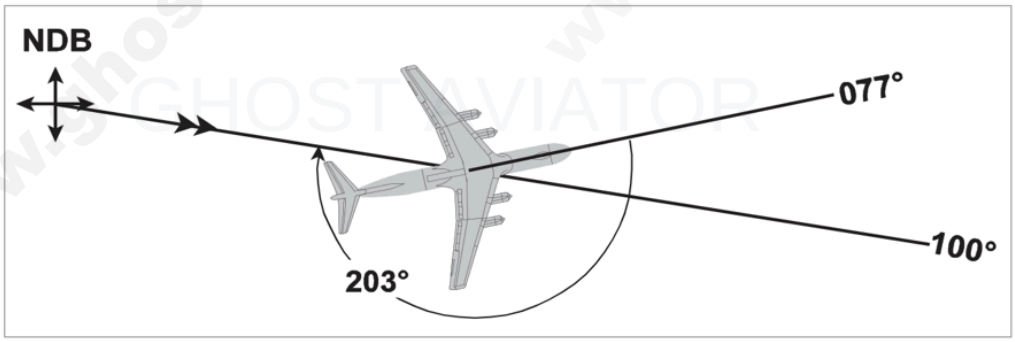


Figure 7.15

Figure 7.14 — Outbound zero wind: heading 260°, RBI 180° (NDB dead astern). Figure 7.15 — 23° starboard drift outbound: heading = track − 23° = 077°, RBI 203°. The same S/P rule applies: Starboard Subtract, Port Plus.

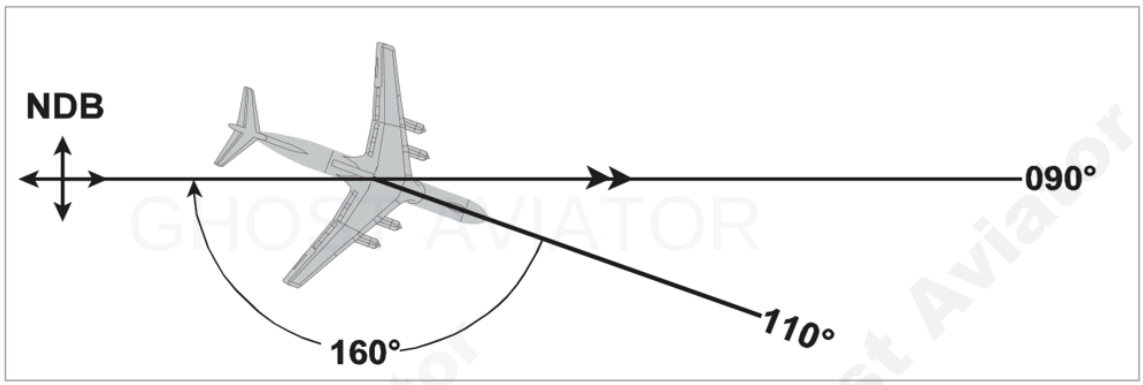


Figure 7.16

Figure 7.16 — 20° port drift outbound: heading = track + 20° = 110°, RBI 160°.

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Track Rule Memory Aid

Starboard drift = **S**ubtract WCA from track heading
Port drift = **P**lus (+) WCA to track heading
(Works for both inbound and outbound)

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10. Drift Assessment

10.1 Inbound

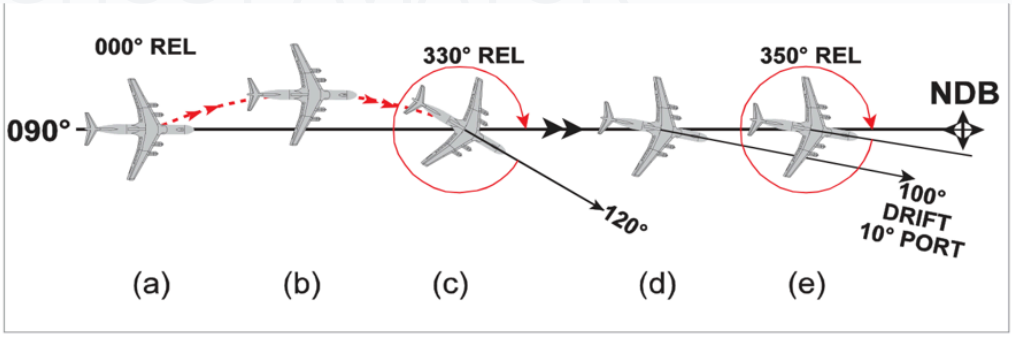


Figure 7.17 Assessing Drift Inbound

Figure 7.17 — Assessing drift inbound: fly with beacon dead ahead (000° RBI). If RBI increases → port drift. Apply 30° starboard turn to regain track. Assess likely drift, apply WCA, and monitor RBI to confirm.

1. Fly the aircraft on the required track with the beacon dead ahead (000° RBI)
2. Maintain heading and watch the RBI:
   - RBI **increases** → port drift (beacon drifting to port)
   - RBI **decreases** → starboard drift
3. Turn say 30° starboard to regain track; RBI will show 330° when back on track
4. Apply estimated drift as WCA; monitor RBI to confirm the assessment is correct

10.2 Outbound

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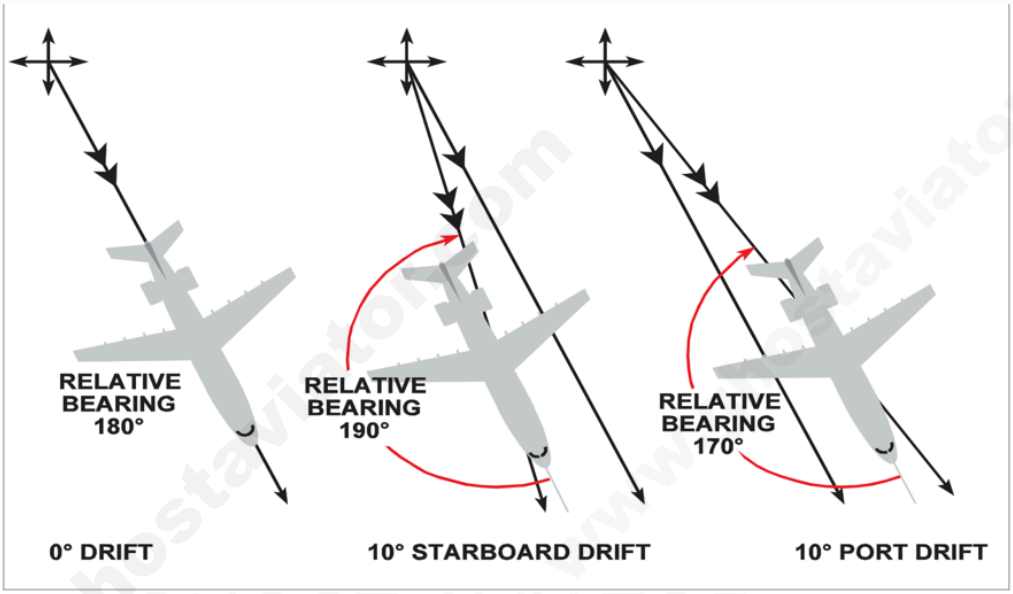


Figure 7.18 Drift Assessing Outbound

In *Figure 7.18* it can be seen that with zero drift the RBI indicates 180° relative. With 10° starboard drift, the relative bearing increases to 190°, and with 10° port drift the relative bearing decreases to 170°. To assess drift by this means the aircraft must maintain a steady heading from directly overhead the beacon.

When the drift has been assessed, alter heading port or starboard, by say 30°, to regain track, until the correct relative bearing of 210° or 150° is obtained. The aircraft is now back on track. The heading must now be altered to take into account the original assessment of drift.

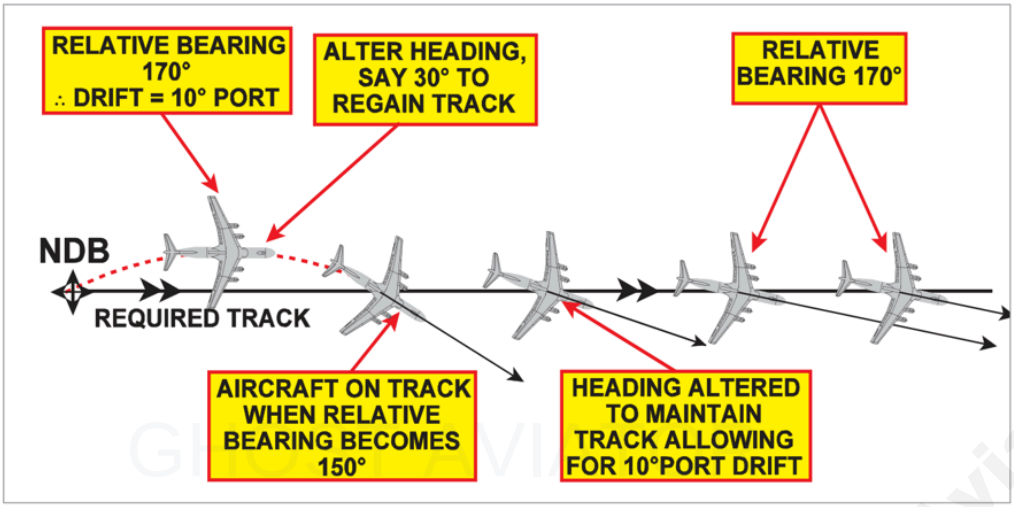


*Figure 7.19 **Deter**mining Drift and Maintaining Track away from an NDB*

Figure 7.18 — Outbound drift: zero drift = RBI 180°; 10° starboard drift = RBI 190°; 10° port drift = RBI 170°. Figure 7.19 — Regain track by turning 30° port or starboard, then apply WCA correction.

- Track directly overhead the beacon first (establishes reference)
- With **starboard drift**: RBI increases from 180° → beacon appears to move to port of astern
- With **port drift**: RBI decreases from 180° → beacon appears to move to starboard of astern
- Turn 30° to regain track, then apply assessed WCA

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11. Holding

When traffic density or weather delays landing, ATC directs aircraft to a **holding area (stack)** organised over a radio beacon. Aircraft fly a racetrack circuit, separated by a minimum of **1000 ft** vertically.

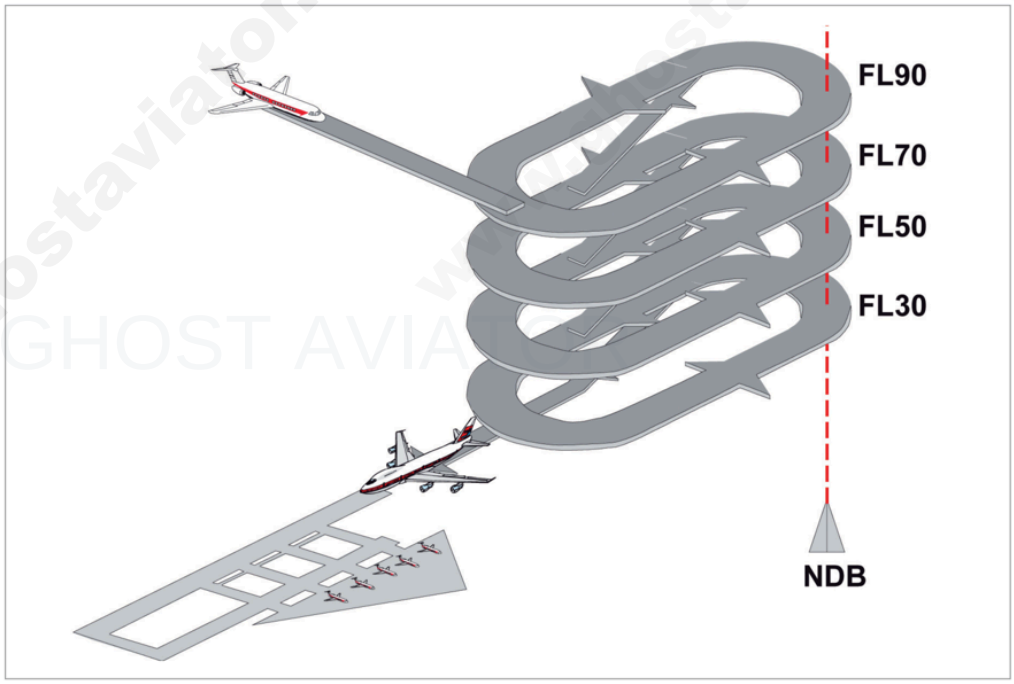


Figure 7.20 The holding system

Figure 7.20 — The holding system: aircraft orbit the NDB in a racetrack pattern, each level separated by 1000 ft minimum. Aircraft descend through the stack when lower levels clear.

12. Runway Instrument Approach Procedures

Most aerodromes have published **NDB instrument approach procedures**. The pilot flies the published procedure to position the aircraft for a visual landing in poor weather. The NDB may be used alone or in conjunction with other approach aids.

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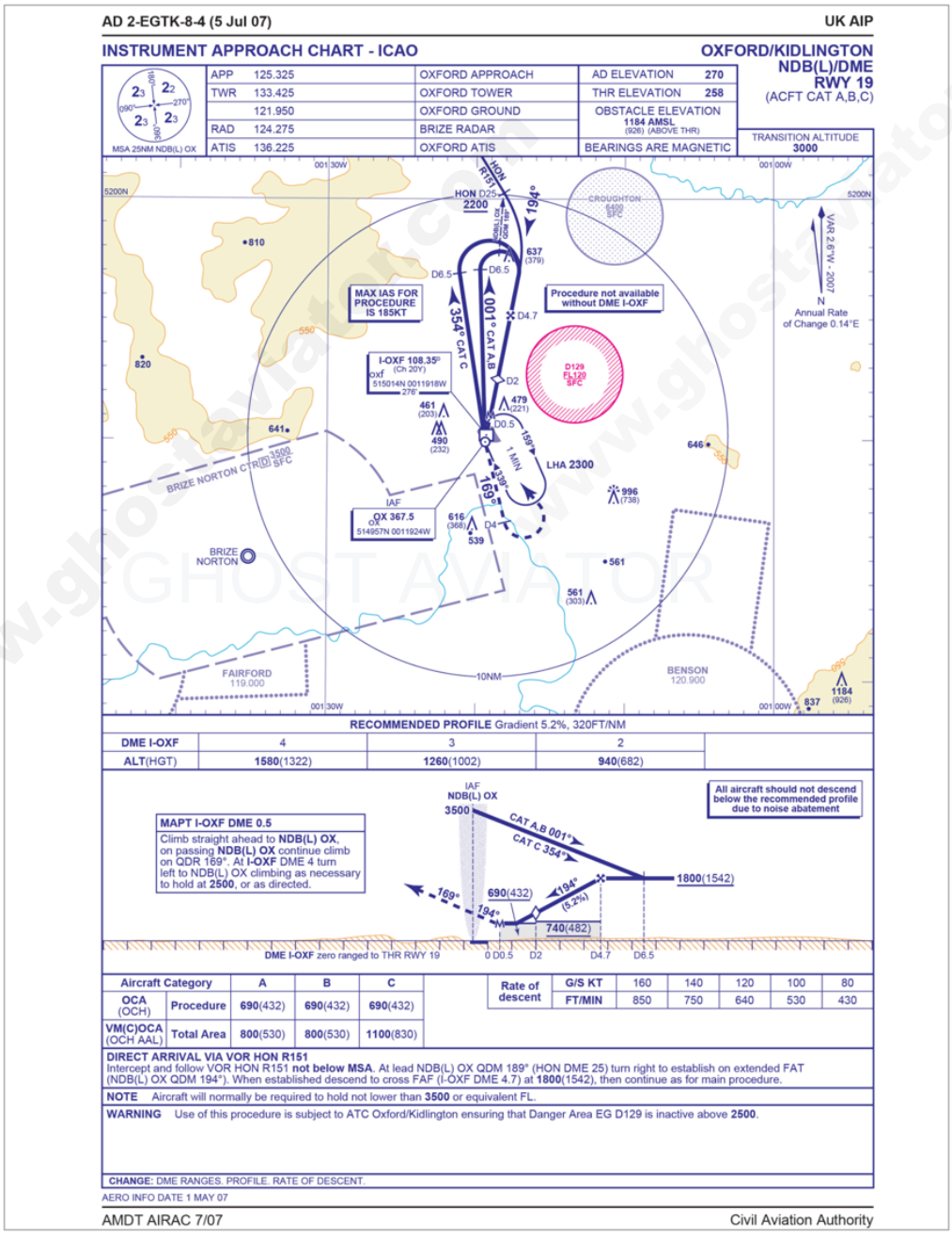


Figure 7.21 Example of an NDB instrument approach

Figure 7.21 — Example NDB instrument approach procedure: shows inbound track, holding pattern, procedure turn, and final approach profile to the runway.

13. Factors Affecting ADF Accuracy

13.1 Designated Operational Coverage (DOC)

The DOC is based on a **daytime protection ratio of 3:1** (signal/noise) between wanted and unwanted signals. Beyond the DOC, bearing errors increase. At night, adverse propagation further increases errors even within the DOC.

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13.2 Static Interference

Type	Cause	Indication
Precipitation static	Collision of water droplets/ice crystals with the aircraft — degrades signal/noise ratio	Wandering needle; background hiss on audio (present on VHF frequencies too)
Thunderstorm static	Powerful static discharges from Cb clouds across the LF/MF spectrum	Loud crackle on audio; needle points rapidly toward the Cb. With multiple active cells, needle may point toward them for prolonged periods.

> **Thunderstorm Practical Note**
>
> *"During Cb activity, the only sensible use of the ADF is to indicate where the active cells are."*

13.3 Night Effect

By day, the **D-region of the ionosphere absorbs LF/MF signals**, preventing sky wave reception. At night, the D-region disappears, allowing **sky wave to contaminate the surface wave**.

- Phase interference between sky wave and surface wave (different path lengths)
- Sky wave's horizontal polarization component induces currents in horizontal loop elements — destroys nulls
- Night effect is **most pronounced at dawn and dusk** (transitional D-region)

13.4 Station Interference

Congestion in LF/MF bands → risk of interference from other stations on or near the same frequency. By day, DOC provides protection. At night, sky wave from out-of-range stations can cause interference *even within the DOC*. **Always positively identify the NDB at night.**

13.5 Mountain Effect

Mountainous terrain causes **reflections and diffraction** of transmitted radio waves → bearing errors. Increases at low altitude; minimised by flying higher.

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13.6 Coastal Refraction

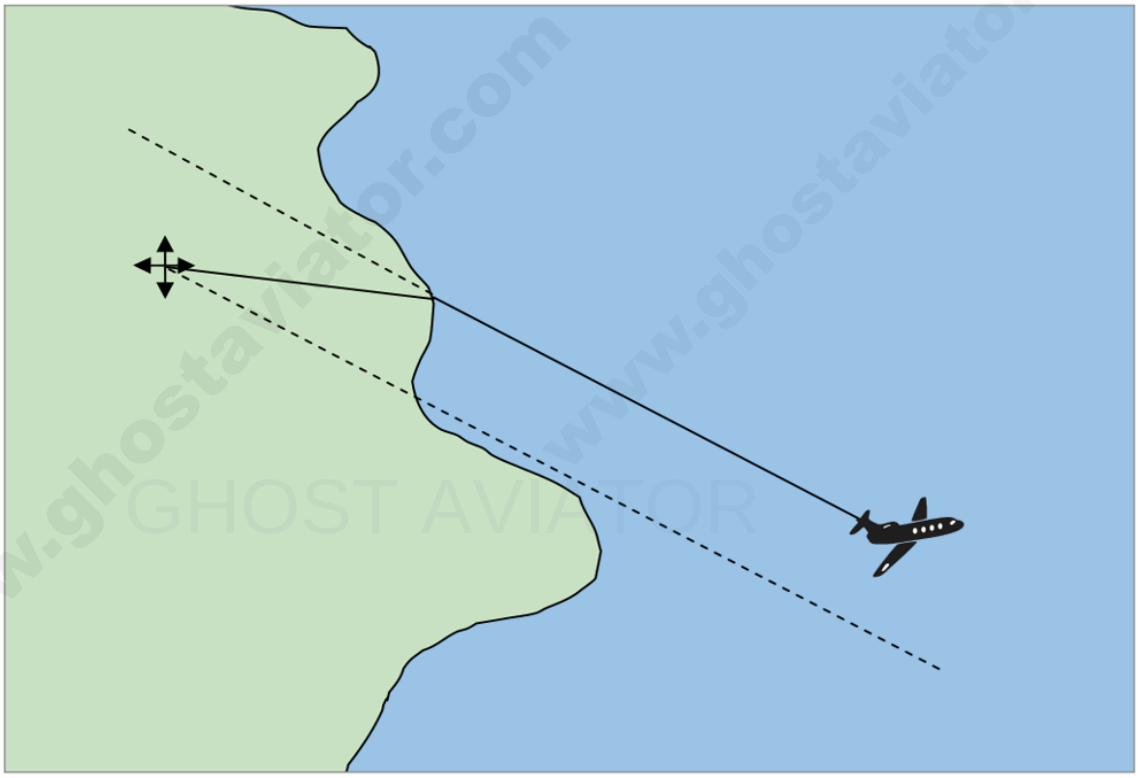


Figure 7.22 Coastal refraction

Figure 7.22 — Coastal refraction: radio waves speed up over water (less attenuation), causing the wavefront to bend away from the normal to the coastline. For an aircraft over the sea, the apparent position is closer to the coast than the actual position.

- Radio waves **speed up over water** (less energy attenuation over sea vs land)
- Speeding up causes the wavefront to **bend away from the normal to the coast** (Snell's law — acceleration bends away)
- Refraction is **negligible at 90°** to the coast; increases as angle of incidence increases
- Aircraft over sea: plotted position appears **closer to the coast** than actual position

Minimising Coastal Refraction

- Use NDBs **on or near the coast**
- Use signals that cross the coast at or **near 90°**
- **Fly higher**

13.7 Quadrantal Error

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The aircraft airframe (predominantly fore-aft axis) distorts the loop aerial's polar diagram. **Incoming NDB signals are refracted toward the fore/aft axis**, with maximum distortion in the quadrants — at relative bearings of **045°, 135°, 225°, 315°**.

- Older ADF systems are regularly "swung" to measure quadrantal error
- Modern systems: error determined by manufacturer; corrections applied electronically

13.8 Angle of Bank (Dip Error)

ADF uses vertically polarized waves. Bank angle tilts the loop, causing its **horizontal elements to pick up the horizontal polarization component** of the incoming wave — inducing circulating currents that destroy the nulls and create bearing errors. This error is **only present when not in level flight**.

13.9 Lack of Failure Warning

Most ADF instruments lack a **failure warning system** — false indications due to equipment failure are not readily detectable. Therefore:

- Always identify and continuously monitor the NDB
- Cross-check with other navaids where possible
- When using ADF as primary navigation aid for an approach, monitor continuously

14. Factors Affecting ADF Range

- **NDB transmission power** — range ∝ √P (see formula above)
- **Frequency** — lower frequencies travel further as surface waves
- **Time of day** — range effectively reduced at night (sky wave contamination)
- **Terrain** — absorbent terrain reduces range; sea extends range (3×√P over water)
- **Receiver sensitivity**

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15. ADF Accuracy and Summary

> **ADF Accuracy**
>
> **±5° within the DOC, by day only.**
> This refers to the measured bearing and does not include compass error.

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NDB Types of NDB: Range (NM):	Ground transmitter in LF or MF band (190 - 1750 kHz) Locator (L) - airfield let-down (10 - 25 NM) En Route - Nav-aid (50 NM or more) $3 \times \sqrt{P}\,(W)$ over water $2 \times \sqrt{P}\,(W)$ over land
ADF	Airborne equipment - aerials, receiver, control unit, indicator (RBI / RMI)
Principle of operation	(Relative) Bearing by switched cardioids
Frequencies	190 - 1750 kHz (LF & MF)
Emission characteristics	N0NA1A - BFO ON for tuning, identification and monitoring N0NA2A - BFO ON for tuning, OFF otherwise
Presentation	RBI or RMI
Uses of NDB	Homing, Holding, Approach, En route nav-aid
Errors	Static interference (precipitation and thunderstorms) Station interference Night effect Mountain effect Coastal refraction Quadrantal error Bank angle (dip) Lack of failure warning
Accuracy (Day Only)	+/- 5° within the DOC

Figure 7.23

Figure 7.23 — ADF Summary: consolidates NDB types, range formulae, airborne equipment, principle, frequencies, emission codes, BFO rules, presentation, uses, errors, and accuracy in a single reference.

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QUICK REVISION — CHAPTER 7

- NDB: ground transmitter, LF/MF, **190–1750 kHz**, vertically polarized, surface wave
- Locator: **10–25 NM** | En route: **50+ NM** | Oceanic: hundreds NM
- Range: 3×√P(W) over water; 2×√P(W) over land
- Loop aerial → figure-8 polar (2 nulls, ambiguous)
- Sense aerial (dipole) → combined with loop → CARDIOID (1 null)
- Switching two cardioids at ~120 Hz → meets ICAO ±5° accuracy
- Goniometer = fixed loop electronic equivalent
- N0NA1A: BFO ON always | N0NA2A: BFO ON for tuning only
- RBI: 000° at nose, add heading to get Mag bearing | RMI: reads Mag directly
- Needle HEAD → QDM (to station) | Needle TAIL → QDR (radial from station)
- Drift rule: Starboard=Subtract; Port=Plus
- Night effect: most pronounced at dusk/dawn
- Coastal refraction: waves speed up over water → bend away from normal; aircraft appears closer to coast
- Quadrantal error max at 045°, 135°, 225°, 315° relative bearings
- Accuracy: **±5° by day within DOC only**
- DOC protection: 3:1 signal/noise ratio by day; NOT guaranteed at night (sky wave)

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Practice Questions & Detailed Answers (21 Questions)

Q1. The phenomenon of coastal refraction which affects the accuracy of ADF bearings:

a. is most marked at night

b. can be minimized by using beacons situated well inland

c. can be minimized by taking bearings where the signal crosses the coastline at right angles

d. is most marked one hour before to one hour after sunrise and sunset

Correct Answer: **(c)**

When the signal crosses the coast at 90°, there is no speed differential component along the wavefront — so no bending occurs. This is the primary mitigation. Option (b) is wrong because inland beacons mean the signal crosses MORE coast/water boundary. Option (a) and (d) describe night effect, not coastal refraction.

Q2. An aircraft is intending to track from NDB 'A' to NDB 'B' on a track of 050° (T), heading 060°(T). If the RBI shows the relative bearing of 'A' to be 180° and the relative bearing of 'B' to be 330° then the aircraft is:

a. port of track and nearer 'A'

b. port of track and nearer 'B'

c. starboard of track and nearer 'A'

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d. starboard of track and nearer 'B'

Correct Answer: **(d)**

Working:
Heading = 060°T. RBI B = 330° → True bearing of B = 060° + 330° − 360° = 030°T
On the correct 050° track, B should bear 050°T from the aircraft (dead ahead of track). B actually bears 030°T — 20° to the LEFT (port) of where expected. This means the aircraft is displaced to the RIGHT (starboard) of the track.
A is at 180° (dead astern, just passed) — B is at 330° rel (30° port of ahead). B being only 30° off dead ahead while A is dead astern confirms the aircraft is closer to B.

The "starboard or port" decision: if the destination beacon is further to the port side than its nominal bearing, you are to the starboard of track. The "nearer" decision: A is behind (180° rel), B is ahead — aircraft is in the second half of the leg, closer to B.

Q3. ADF quadrantal error is caused by:

a. static build up on the airframe and St. Elmo's Fire

b. the aircraft's major electrical axis, the fuselage, reflecting and re-radiating the incoming NDB transmissions

c. station interference and/or night effect

d. NDB signals speeding up and bending as they cross from a land to water propagation path

Correct Answer: **(b)**

Quadrantal error is caused by the fuselage (major fore/aft electrical axis) distorting the ADF's polar diagram by reflecting and re-radiating incoming

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signals. Error is maximum at 045°, 135°, 225°, 315° relative bearings (in the quadrants). Option (d) describes coastal refraction.

Q4. The overall accuracy of ADF bearings by day within the promulgated range (DOC) is:

a. ± 3°

b. ± 5°

c. ± 6°

d. ± 10°

Correct Answer: **(b)** — ±5°

ADF accuracy = ±5° within DOC by day. Note this is the *bearing* accuracy only and does not include any additional compass error.

Q5. In order to Tune, Identify and Monitor N0NA1A NDB emissions the BFO should be used as follows:

a. Tune ON / Identify ON / Monitor OFF

b. Tune ON / Identify ON / Monitor ON

c. Tune ON / Identify OFF / Monitor OFF

d. Tune OFF / Identify OFF / Monitor OFF

Correct Answer: **(b)** — BFO ON for all three

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N0NA1A = interrupted unmodulated carrier. An unmodulated carrier cannot be heard without a BFO. The BFO creates an audible tone by mixing the carrier with an offset frequency. Therefore BFO must remain ON during all phases: tuning, identification, and monitoring. Option (a) is for N0NA2A (where monitoring can be done with BFO off).

Q6. The magnitude of the error in position lines derived from ADF bearings that are affected by coastal refraction may be reduced by:

a. selecting beacons situated well inland

b. only using beacons within the designated operational coverage

c. choosing N0NA2A beacons

d. choosing beacons on or near the coast

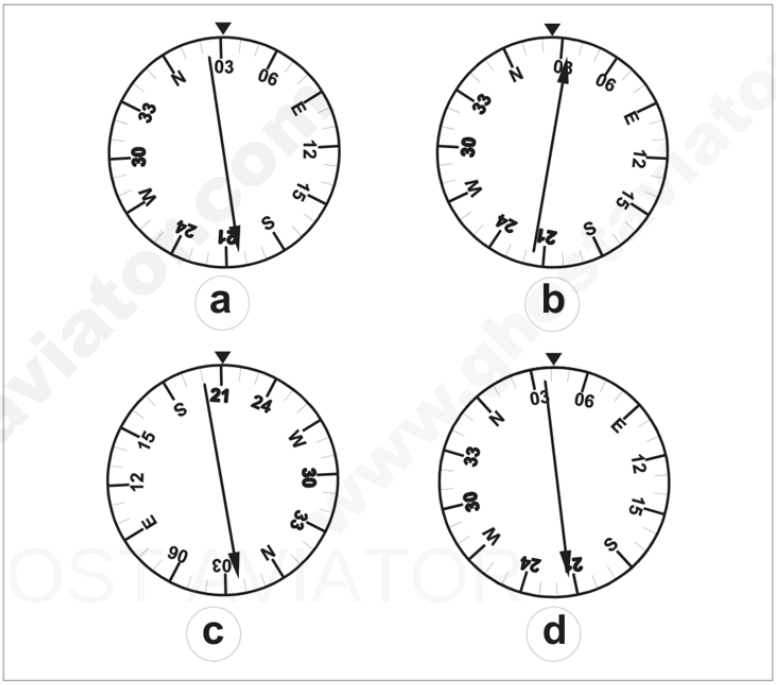
Correct Answer: **(d)**

Beacons on or near the coast minimise the over-water propagation path — so the signal spends less distance over the sea before crossing the coast, reducing refraction. Option (a) — inland beacons increase the refraction. Option (b) — DOC applies to signal quality, not coastal geometry. Option (c) — emission type has no effect on refraction.

Q7. An aircraft is tracking away from an NDB on a track of 023°(T). Drift is 8° port, variation 10° west. Which of the RMIs illustrated below shows the correct indications?

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indications?



8. **The BFO facility on ADF equipment should be used as follows when an NDB having N0NA1A type emission is to be used:**

a. BFO on for tuning and identification but may be turned off for monitoring
b. BFO on for tuning but can be turned off for monitoring and identification purpose
c. BFO off during tuning, identification and monitoring because this type of emission is not modulated
d. BFO should be switched on for tuning, ident and monitoring

9. **The protection ratio of 3:1 that is provided within the promulgated range/designated operational coverage of an NDB by day cannot be guaranteed at night because of:**

a. long range sky wave interference from other transmitters
b. sky wave signals from the NDB to which you are tuned
c. the increased skip distance that occurs at night
d. the possibility of sporadic E returns occurring at night

Correct Answer: **(d)**

Working:

Track outbound = 023°T. Port drift → heading = track + WCA = 023° + 8° = 031°T

Variation 10°W → Magnetic heading = 031° + 10° = 041°M

NDB is behind (outbound). QDR from NDB = track direction magnetic = 023° + 10° = 033°M

Bearing TO NDB (QDM) = 033° + 180° = 213°M (needle head on RMI)

RMI shows: compass at 041°M at top; needle pointing to 213°M (roughly astern, slightly to port)

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Key steps: (1) apply drift to get heading; (2) apply variation to get magnetic; (3) NDB bears reciprocal of outbound QDR from aircraft; (4) check which RMI diagram matches heading 041°M with needle at ~213°M.

Q8. The BFO facility on ADF equipment should be used as follows when an NDB having N0NA1A type emission is to be used:

a. BFO on for tuning and identification but may be turned off for monitoring

b. BFO on for tuning but can be turned off for monitoring and identification

c. BFO off during tuning, identification and monitoring because this type of emission is not modulated

d. BFO should be switched on for tuning, ident and monitoring

Correct Answer: **(d)**

N0NA1A contains an interrupted unmodulated carrier (A1A) which requires the BFO to produce any audible signal. BFO must remain ON throughout — tuning, identification, and continuous monitoring. This is identical to Q5 but phrased differently.

Q9. The protection ratio of 3:1 that is provided within the promulgated range/DOC of an NDB by day cannot be guaranteed at night because of:

a. long range sky wave interference from other transmitters

b. sky wave signals from the NDB to which you are tuned

c. the increased skip distance that occurs at night

d. the possibility of sporadic E returns occurring at night

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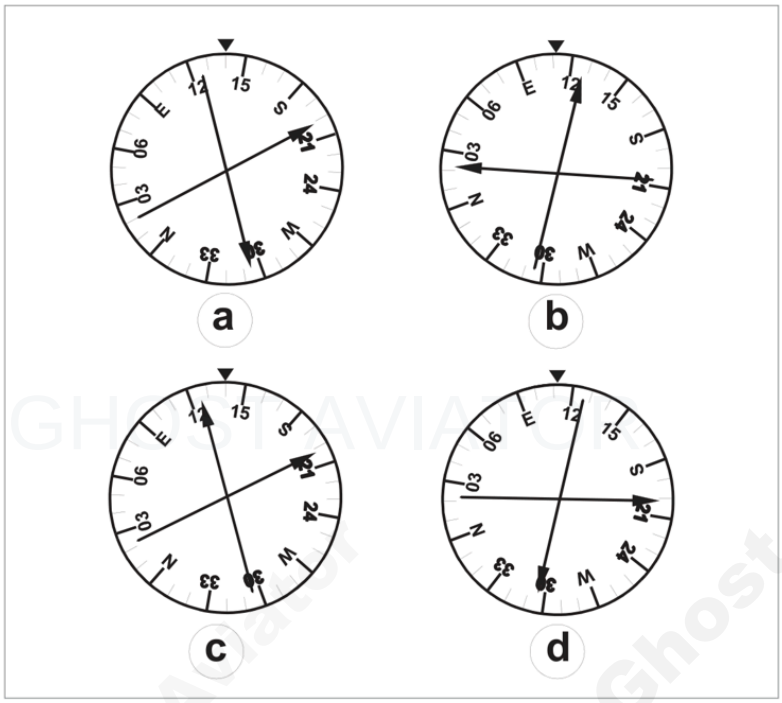
Correct Answer: **(a)**

At night, the D-region disappears — distant NDBs on similar frequencies can propagate via sky wave over thousands of miles and cause interference even within another NDB's DOC. The "wanted" NDB's signal/noise ratio drops below 3:1 because of these long-range sky wave intruders. Option (b) is incorrect — it's other stations, not the tuned NDB, that cause night interference.

Q10. An aircraft has an RMI with two needles. Assume that: (i) The aircraft is outbound from NDB Y on a track of 126°(M), drift is 14° port; (ii) A position report is required when crossing a QDR of 022 from NDB Z. Which diagram represents the RMI at the time of crossing the reporting point?

i) The aircraft is outbound from NDB Y on a track of 126 (M) drift is 14° Port
ii) **A position report is required when crossing a QDR of 022 from NDB Z**

Which of the diagrams below represents the RMI at the time of crossing the reporting point?



11. **Each NDB has a range promulgated in the COMM section of the AIP. Within this range interference from other NDBs should not cause bearing errors in excess of:**

a. day ± 5°
b. night ± 10°

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Correct Answer: **(a)**

Working:
Outbound track from Y = 126°M. Port drift 14° → heading = 126° + 14° = 140°M
Needle Y: NDB Y is behind → points 126° + 180° = 306°M (outbound reciprocal)
QDR from Z = 022°M → position report when crossing this radial → QDM to Z = 022° + 180° = 202°M
Needle Z: points to 202°M
RMI: heading 140°M on compass card; Y needle at 306°M; Z needle at 202°M → matches option (a)

Q11. Each NDB has a range promulgated in the COMM section of the AIP. Within this range interference from other NDBs should not cause bearing errors in excess of:

a. day ±5°

b. night ±10°

c. day ±6°

d. night ±5°

Correct Answer: **(a)** — day ±5°

Within the promulgated DOC, the 3:1 protection ratio is guaranteed by day, ensuring bearing errors stay within ±5°. The protection is *not* guaranteed at night due to sky wave interference from distant stations.

Q12. The range promulgated in the AIP and flight guides for all NDBs in the UK is the range:

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a. within which a protection ratio of 3:1 is guaranteed by day and night

b. up to which bearings can be obtained on 95% of occasions

c. within which bearings obtained by day should be accurate to within 5°

d. within which protection from sky wave protection is guaranteed

Correct Answer: **(c)**

The DOC defines the range within which (by day) the 3:1 protection ratio ensures ±5° accuracy. Options (a) and (d) are wrong — protection is NOT guaranteed at night. Option (b) is an invented distractor not matching the DOC definition.

Q13. In order to resolve the 180° directional ambiguity of a directional LOOP aerial its polar diagram is combined with that of a SENSE aerial _____ to produce a _____ whose single null ensures the ADF needle moves the shortest distance to indicate the correct _____:

a. at the aircraft / cardioid / radial

b. at the transmitter / limacon / bearing

c. at the aircraft / limacon / bearing

d. at the aircraft / cardioid / bearing

Correct Answer: **(d)**

The sense aerial is combined with the loop aerial *at the aircraft* (within the goniometer). The combined diagram is a CARDIOID. The cardioid's single null ensures the needle takes the shortest path to indicate the correct BEARING. Option (a) is wrong — the output is a bearing, not a radial (radials are from a

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ground station). Options (b) and (c) are wrong — the combined diagram is a cardioid, not a limacon.

Q14. The protection ratio afforded to NDBs in the UK within the promulgated range (DOC) applies:

a. by day only

b. by night only

c. both day and night

d. at dawn and dusk

Correct Answer: **(a)** — by day only

The 3:1 protection ratio is based on daytime conditions when the D-region absorbs sky wave from distant stations. At night, the D-region disappears — sky wave propagation from distant co-channel stations renders the protection ratio meaningless within the DOC.

Q15. The phenomenon of coastal refraction affecting ADF bearings is caused by the signal _____ when it reaches the coastline and bending _____ the normal to the coast:

a. accelerating / towards

b. decelerating / towards

c. accelerating / away from

d. decelerating / away from

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Correct Answer: **(c)** — accelerating, away from

Radio waves travel faster over water (lower attenuation). When a signal crosses the coast from land to sea, it **accelerates**. By Snell's law, acceleration causes the wavefront to bend **away from** the normal (perpendicular) to the coast. This is the opposite of light entering a denser medium. Options (b) and (d) would describe deceleration (entering a denser medium).

Q16. In an ADF system, night effect is most pronounced:

a. during long winter nights

b. when the aircraft is at low altitude

c. when the aircraft is at high altitude

d. at dusk and dawn

Correct Answer: **(d)** — at dusk and dawn

At dusk and dawn the D-region is in transition — partially present. The mixture of surface wave and sky wave at this time creates maximum phase interference, producing the worst bearing errors. The term "night effect" is somewhat misleading — it peaks at dusk/dawn, not during the middle of the night.

Q17. When the induced signals from the loop and the sense antenna are combined in an ADF receiver, the resultant polar diagram is:

a. a limacon

b. a cardioid

c. figure of eight shaped

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d. circular

Correct Answer: **(b)** — a cardioid

Loop (figure-8) + sense aerial (circle) = CARDIOID when the sense aerial signal has equal amplitude to the loop signal. A cardioid has a single null, resolving the 180° ambiguity. Option (c) is the loop alone; option (d) is the sense aerial alone.

Q18. When flying over the sea and using an inland NDB to fix position with a series of position lines, the plotted position in relation to the aircraft's actual position will be:

a. further from the coast

b. closer to the coast

c. co-incident

d. inaccurate due to the transmitted wave front decelerating

Correct Answer: **(b)** — closer to the coast

The signal from the inland NDB crosses the coast and travels over the sea toward the aircraft. Over the sea it speeds up and bends away from the coast's normal — making the signal appear to come from a direction slightly further inland than actual. The aircraft, applying this bearing to fix position, will plot itself closer to the coast than its actual position. Option (d) is wrong — the wave accelerates, it does not decelerate.

Q19. An aircraft on a heading of 235°(M) shows an RMI reading of 090° with respect to an NDB. Any quadrantal error which is affecting the accuracy of this

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bearing is likely to be:

a. a maximum value

b. a very small value

c. zero, since quadrantal error affects only the RBI

d. zero, since quadrantal error affects only the VOR

Correct Answer: **(a)** — a maximum value

Working:
NDB magnetic bearing = 090°M (from RMI). Aircraft heading = 235°M.
Relative bearing = 090° − 235° + 360° = 215°
Quadrantal error is maximum at relative bearings of 045°, 135°, 225°, 315°.
215° is close to 225° — near a maximum quadrantal error position. So the error is near maximum.
Options (c) and (d) are completely wrong — quadrantal error affects ALL ADF systems using a loop aerial, including both RBI and RMI.

Q20. The principal propagation path employed in an NDB/ADF system is:

a. sky wave

b. surface wave

c. direct wave

d. ducted wave

Correct Answer: **(b)** — surface wave

NDBs operate in LF/MF bands where surface wave propagation is the primary mode. Surface waves follow the earth's curvature and provide reliable, predictable

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bearings. Sky wave exists at night but causes errors (night effect) — it is an undesired mode, not the intended propagation path.

Q21. The ADF of an aircraft on a heading of 189°(T) will experience the greatest effect due to quadrantal error if the NDB bears:

a. 234°(T)

b. 279°(T)

c. 225°(T)

d. 145°(T)

Correct Answer: **(a)** — 234°(T)

Working:

Quadrantal error is maximum at relative bearings 045°, 135°, 225°, 315°.

Aircraft heading 189°T. For each maximum error relative bearing:

— 189° + 045° = 234°T ✓ (matches option a)

— 189° + 135° = 324°T

— 189° + 225° = 054°T (414° − 360°)

— 189° + 315° = 144°T

Only 234°T appears in the options → answer is (a).

Note: option (c) 225°T gives relative bearing = 225° − 189° = 036° — not a maximum error position.

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Master Reference Tables

BFO Quick Reference

Emission	Tune	Identify	Monitor
N0NA1A	ON	ON	ON
N0NA2A	ON	OFF	OFF

ADF Errors Quick Reference

Error	Cause	Mitigation
Static (precipitation)	Water/ice collision with airframe	None; wait for conditions to clear
Static (thunderstorm)	Cb electrical discharges	Avoid Cb; treat needle as Cb pointer
Night effect	D-region disappears; sky wave contamination	Use within DOC; identify + monitor; worst at dawn/dusk
Station interference	Co-channel stations via sky wave at night	Use within DOC by day; always identify
Mountain effect	Terrain reflections and diffraction	Fly higher
Coastal refraction	Waves accelerate over sea → bend from normal	NDBs on coast; cross at 90°; fly higher
Quadrantal error	Fuselage re-radiates signal; max at 045/135/225/315° rel	Manufacturer correction; swing compass
Dip (bank) error	Loop tilt in turns induces horizontal currents	Only fly ADF procedures in level flight
Failure warning absent	No flag/warning on most ADF instruments	Continuously monitor and cross-check

Quadrantal Error — Maximum Positions

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Relative Bearing	Error
045° / 135° / 225° / 315°	MAXIMUM
000° / 090° / 180° / 270°	Minimum (zero)

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